

Gregorio Zaltzman D'Ambrosio

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PROFILE

Trilingual Aerospace Engineering senior and dual EU (Italy)/Argentine citizen. Culturally adaptable, having lived in six countries. Combines a rigorous analytical background with hands-on leadership in systems integration, complex aircraft design, and launch vehicle development.

EDUCATION

University of California, San Diego

San Diego, CA | Sep 2022 – Jun 2026

B.S. Aerospace Engineering | GPA: 3.2

Coursework: Aerodynamics, Aerospace Structures, Propulsion, Orbital Mechanics, Linear Control, Fluid Dynamics, Thermodynamics, Solid Mechanics, Numerical Methods, Engineering Experimental Techniques, Signals & Systems

British School of Paris

Paris, France | Jun 2022

A-Levels: A* Mathematics, A Physics, A* EPQ | SAT: 780M / 660E | Duke of Edinburgh Gold Award

SKILLS

Analysis & Simulation MATLAB, Python, XFLR5, XFOIL, AVL, OpenVSP, STK, FEA, CFD

Design & CAD SolidWorks, Fusion 360, technical drawings & assemblies

Hardware & Lab Arduino IDE, sensor instrumentation, data logging & post-processing, rapid prototyping

Languages English, Spanish, French (all fluent)

ENGINEERING PROJECTS

BWB Commercial Airliner “Manta” — Chief Engineer

Jan 2026 – Present

- Led a 6-person team designing a 250-PAX Blended Wing Body for Mach 0.85 and 7,400 nmi.
- Maximized cruise L/D to 25.30 via Python Particle Swarm Optimization under yield and static stability constraints.
- Conducted 500+ Latin Hypercube Monte Carlo simulations to identify aspect ratio as the primary L/D driver.
- Engineered directional stability without a vertical stabilizer via Fly-By-Wire control and optimized winglets.
- Integrated aft twin engines for acoustic shielding and sized a PRSEUS fuselage, achieving \$0.0873 CASM.

Wing Spar Structural Analysis — MATLAB Tool

Feb 2025

- Built MATLAB tool using thin-walled beam theory to analyze a single-cell wingbox under combined aerodynamic loading.
- Conducted Carbon/Epoxy vs. Al-7075-T6 trade study showing substantial mass savings; generated full stress/twist plots.

Clark Y-14 Wind Tunnel Aerodynamic Validation

March 2026

- Characterized lift/drag/moment across pre- and post-stall AoA via pressure taps, wake surveys, and sting balances.
- Identified $\approx 2^\circ$ discrepancy in separation onset versus XFLR5 models, applying empirical corrections to BWB estimates.

SeaGlide Autonomous Underwater Glider — Systems Integration

Feb 2026

- Designed and built a buoyancy-driven AUV with 3D-printed structure, Arduino control, and motorized syringe.
- Validated MATLAB glide path simulations ($\pm 44.5^\circ$) against physical pool trials to verify buoyancy performance.

EXPERIENCE

Design-Build-Fly (DBF) — Aerodynamics & Structures

UC San Diego | Sep 2025 – Present

- Executed airfoil selection and configuration optimization for a competition RC aircraft via performance-driven trade studies.
- Conducted structural sizing, engineering analysis, and CAD modeling in collaboration with interdisciplinary teams.

Rocket Propulsion Laboratory (RPL) — Daedalus Rocket

UC San Diego | Jun 2025 – Present

- Supported the design, assembly, and launch of a G-class model rocket, achieving a targeted apogee of $\sim 2,100$ ft.
- Performed flight performance analysis in MATLAB and contributed to CAD, avionics integration, and stability testing.

ITS Service Desk — Lead Technician

UC San Diego | Sep 2023 – Present

- Directed a 60+ member technician team delivering Tier-1 IT support and technical troubleshooting for the university.
- Managed daily operations and escalations; founded a zero-tier support social media channel to streamline issue resolution.

C2N @ CNRS — Intern

Paris, France | Jul 2022

- Analyzed nanophotonic crystals and nanolasers; operated an electron microscope and performed clean room fabrication.

LEADERSHIP & ACTIVITIES

- **International Student Society** — Event Coordinator & Recruitment Chair: mentored 50+ international students.
- **Phi Gamma Delta** — Scholarship & Graduate Relations Chair: organised professional development and networking events.
- **Duke of Edinburgh Gold Award** — Sustained leadership, community service, and expedition (British School of Paris).